



Cambridge (CIE) IGCSE Computer Science



Your notes

Development Life Cycle

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- * Program Development Life Cycle



Your notes

Program Development Life Cycle

Analysis

What is the analysis stage of the program development life cycle?

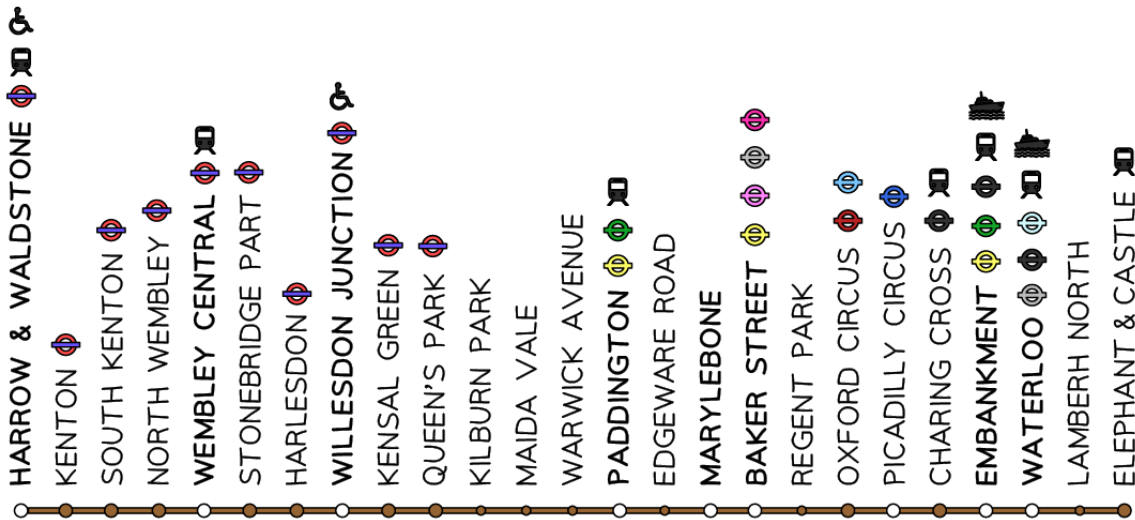
- The analysis stage of the program development life cycle is to **precisely understand the problem** the program is intended to solve

What role does abstraction play in the analysis?

- Abstraction is the act of **removing unimportant details** from the problem to **focus on important elements** such as:
 - **Core functionality**
 - **Requirements**
- An example of abstraction would be the London underground train route map; travellers do not need to know the geographical layout of the routes, only that getting on at stop A will eventually transport you to stop B



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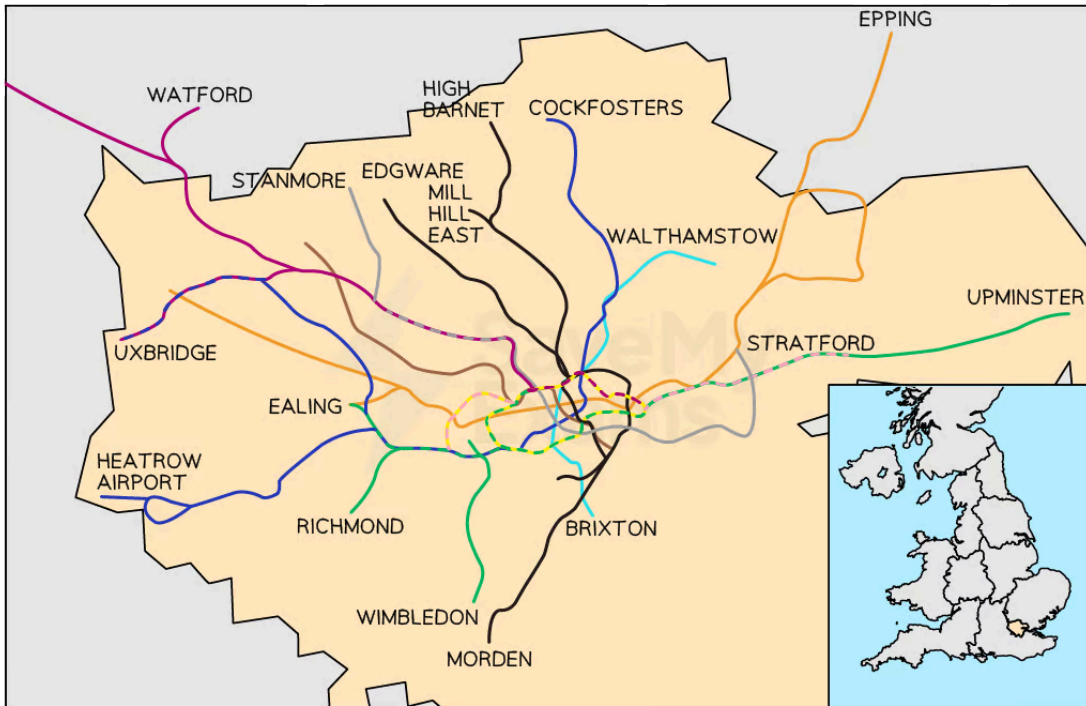
KEY:

-  INTERCHANGE STATION
-  MAIN STATION
-  MINOR STATION
-  CONNECTION TO LONDON OVERGROUND SERVICE
-  CONNECTION TO CIRCLE LINE SERVICES
-  CONNECTION TO CENTRAL LINE SERVICES
-  CONNECTION TO DISTRICT LINE SERVICES
-  CONNECTION TO HAMMERSMITH & CITY LINE SERVICES
-  CONNECTION TO JUBILEE LINE SERVICES
-  CONNECTION TO METROPOLITAN LINE SERVICES
-  CONNECTION TO NORTHERN LINE SERVICES
-  CONNECTION TO PICADILLY LINE SERVICES
-  CONNECTION TO VICTORIA LINE SERVICES
-  CONNECTION TO WATERLOO & CITY LINE SERVICES
-  CONNECTION TO NATIONAL RAIL SERVICES
-  CONNECTION TO LONDON RIVER SERVICES
-  WHEELCHAIR ACCESS

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Core functionality

- Using abstraction helps to **identify the fundamental components** of what the program is going to solve
- Before tackling a problem, it needs to be clearly **understood by everyone working on it**
- The overall goal of the solution needs to be agreed as well as **any constraints** such as limited resources or requiring a **platform** specific solution

Requirements

- To create a solution, **a requirements document is created** to define the problem and break it down into clear, manageable, understandable parts by using abstraction and decomposition
- A requirements document **labels each requirement**, gives it a description as well as **success criteria** which state how we know when the requirement has been achieved

Design

What is the design stage of the program development life cycle?



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- The design stage of the program development life cycle involves using **techniques** to come up with a **blueprint for a solution**
- Ways the design of a solution to a problem can be presented include:
 - **Structure diagrams**
 - **Flowchart**
 - **Pseudocode**

Coding

What is the coding stage of the program development life cycle?

- Developers begin programming **modules** in a **suitable programming language** that works together to provide an overall solution to the problem
- As each developer programs, they **perform iterative testing**
 - Iterative testing is where each module is tested and debugged thoroughly to make sure it interacts correctly with other modules and accepts data without crashing or causing any errors
- Developers may need to retest modules as new modules are created and changed to make sure they continue to interact correctly and do not cause errors

Testing

What is the testing stage of the program development life cycle?

- Once the overall program or set of programs is created, they are run many times using varying sets of test data
- This ensures the program or programs work as intended as outlined in the initial requirements specification and design and rejects any invalid data that is input
- Examples of test data include **alphanumeric** sequences to test password validation routines